

Investigating the Impact of Actuator Airfoil Design on Transportation Costs in UAV Morphing Wing Systems

Saad Farooq, saad.farooqf7@gmail.com

ACKNOWLEDGEMENTS

Thank you for the guidance of Jose Rivas from in the development of this research paper.

Abstract —

Morphing wing technology allows Unmanned Aerial Vehicles (UAVs) to change the shape of their wings during flight to reduce the effect of changing flight conditions on performance. This Paper studies the impact that airfoil shape and actuator technology have on the aerodynamic efficiency and cost of transportation in UAV morphing wing systems. In this research case, comprehensive actuator modeling, airfoil comparisons and performance-cost analysis are used in determining the optimal airfoil-actuator pairs that yield the most efficient results at different phases of flight. Experimental actuator information and aerodynamic coefficients would allow researchers to model the high-lift to low-drag configuration shifts to determine energy savings and cost alterations. The most appropriate is the electric linear actuators that give the switch of NACA 9412 to NACA 0012 that brings in notable changes in flight endurance as well as cost benefit in operations.

I. INTRODUCTION

The growing diversity and commercial use of unmanned aerial vehicles (UAVs) in logistics and surveillance missions demand greater efficiency in the flight performance in various mission configurations. The conventional fixed-wing UAVs perform optimally only in a certain flight regime because they have to trade between lift generation (during takeoff) and drag (during cruise). The morphing wing technology allows UAVs to change the shape of wings in-flight that leads to the optimization of performance due to real-time adjustments.

These benefits of morphing do not occur without cost. Morphing energy requirements and mechanical complexity plus the weight of actuators tend to reduce the overall system advantage. An extensive test should be conducted to determine the most effective airfoil-actuator combinations that lead to benefit in energy consumption and cost of operation and long-term flight time.

This paper investigates the influence of wing geometry design and actuator selection on the aerodynamic performance and transportation cost in morphing wing UAVs. Our actuator power requirements and NACA airfoil aerodynamic features and propulsion energy advantages coupled analysis enables us to understand the best combinations that surpass the traditional fixed-wing aircraft in both effectiveness of operation and cost-effectiveness. It also Unlike [10], which isolated aerodynamic gain, our study integrates actuator energy and lifecycle cost modeling. This work also uniquely combines actuator data with aerodynamic modeling to quantify lifecycle cost savings.

Section II explains the procedures for testing airfoils together with their associated actuators. Section III examines different types of actuators. Section IV examines how different airfoil designs impact aerodynamic efficiency. Section V presents a transportation cost model. Section VI integrates results to recommend an optimal configuration. Section VII discusses the broader implications. Section VIII concludes the paper.

Index Terms

Airfoil Performance, Electric Actuator, Morphing Wing, UAV Transportation Cost

II. LITERATURE REVIEW

Having introduced the morphing concept, the next section examines prior research on actuator design and aerodynamic shaping. The ability of morphing wing systems to enhance the UAV operational capabilities has been well researched in studies. In [9], the researchers demonstrated using Bayesian optimization that drones capable of wing and tail morphing could fly with 11.5 % more energy efficiency when they adapt bird-like flight behaviors. The system demonstrated capability of sustaining stability during turbulent air and actuator failures because it used all the control variables.

[10] conducted a comparison of the performance of span and camber morphing on a 25-kg UAV. Researchers found that span morphing offered the greatest increases in aerodynamic efficiency of 25 per cent particularly when used in combination with retractable wing systems at high speeds when fast flight was involved. Aerodynamic advantages of camber morphing achieved 4.5 percent and also its performance response turned out to be more sensitive as the flight speeds escalated. The team discovered that the span morphing had a superior resistance during the varied mission operation but needed sophisticated structure modifications.

[1] and [2] have researched on the basic elements of morphing mechanisms that are; compliant skins and variable-camber and telescopic structures and shape memory alloys. The basic research established the basis of investigating morphing systems that are capable of attaining structural efficiency alongside aerodynamic performance.

Most of the existing work, however, is in either aerodynamic performance or control dynamics, despite these encouraging signs. Little firm work has been done on altogether quantifying transport costs, especially with realistic actuator energy consumption, drag decrease measurements, and lifetime mission investigation. The gap is filled in this paper, which integrates actuator data and airfoil performance with a mission-driven cost model to develop engineering understanding of the cost-effectiveness of UAV morphing wing systems.

III. METHODOLOGY

Bringing together aerodynamic modeling, analysis of actuator performance, and estimation of transportation cost accommodates this study to investigate the performance of different actuator airfoil configurations in morphing wing UAVs.

III.A Selection of airfoils and Aerodynamic simulations

We have chosen four representative 0012, 2412, 4412 and 9412- based on published lift and drag behavior versus angle of attack. Symmetrical airfoils such as the NACA 0012 are of low-drag and suitable in cruise modes and cambered airfoils such as the NACA 9412 provide high lift in takeoff and slow-flight modes. Published airfoil data, usually obtained in wind tunnel tests (e.g., Selig et al. [6]), were used to obtain lift-to-drag (Cl/Cd) ratios at representative UAV flight conditions. XFLR5 was used to for simulations of the flight phase transitions and energy trade-offs modeled airfoil transitions.

III.B Performance and specification data Actuator

We compared technical specifications and datasheets of various servo motors and electric linear actuators. The most important measures were the stroke length, stall force, power draw, weight and responsiveness of the controls. These were standardized to permit comparison and were considered in terms of the way that they might be used to provide actuating wing deformations such as trailing edge deflection or surface camber alteration.

III.C Modeling of Transportation Costs

A UAV flight was characterized by a 2-hour flight profile that consisted of three phases, i.e., high-lift takeoff, low-drag cruise, and stability and control descent. The consumption of energy was estimated in:

Actuator displacement vs transition

- Airfoil drag coefficient propulsion power
- Actuator-added mass effects propulsion

- Cumulative mission-level energy use and cost

We have assumed an average cost of energy of \$0.20 per kWh, and estimated the energy saved per mission and extended this to a 10,000-mission lifetime. The 2-hour mission profile reflects typical UAV logistics operations, and \$0.20/kWh aligns with commercial U.S. energy rates. By using this model, the actuator--airfoil strategies can be compared with each other based on their relative contribution to the global energy and cost efficiency.

IV. ACTUATOR TECHNOLOGIES USED IN UAV MORPHING WINGS

The selection of actuator is crucial to the performance and feasibility of a morphing wing system. There are two actuator solutions high under consideration which are electric linear actuators and servo motors.

A. Servo Motors

Servo motors are small, light, and typically angular displacement applications (flap or rudder) are best suited. Servos are good enough in precision but not applicable to actuate large airfoil deformations or larger morphing skins due to their low stall torque and small linear range. Servos also increase current consumption when loaded, resulting in poor energy consumption during long missions or cycling morphing operations.

MODEL	STROKE (°)	VOLTAGE (V)	IDLE POWER (W)	STALL POWER (W)	TORQUE (OZ·IN)	WEIGHT (G)
HS-7955TG	180	4.8–7.4	1.44–2.22	20.16–31.08	250–402	65
D141SH	180	3.5–8.4	0.56–1.344	8.75–21	97.21	26
HS-7115TH	180	6–7.4	0.78–0.962	9–11.1	44–54	20
D645MW	180	4.8–7.4	2.4–3.7	12.72–19.61	115–179.15	57
SB-2290SG	120	6–7.4	1.2–1.48	50.1–61.79	625–972.1	81
D131MW	180	7.4	0.222	6.66	40	9.8
Savox SV-1260MG	120	6–7.4	1.08–1.332	28.2–34.78	111.1–166.6	40
Savox SC-0252MG+	120	4.8–6.0	1.584–1.98	8.16–10.2	99.9–145.8	49

TABLE I
TABLE OF SERVO MOTORS AND THEIR SPECIFICATIONS

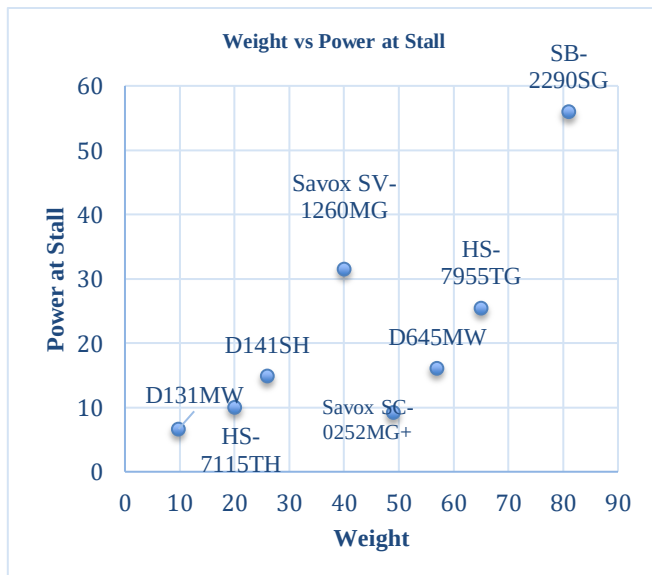


FIGURE I
PLOT OF RELATIONSHIP OF THE WEIGHT OF
SERVOS AND THEIR POWER AT STALL

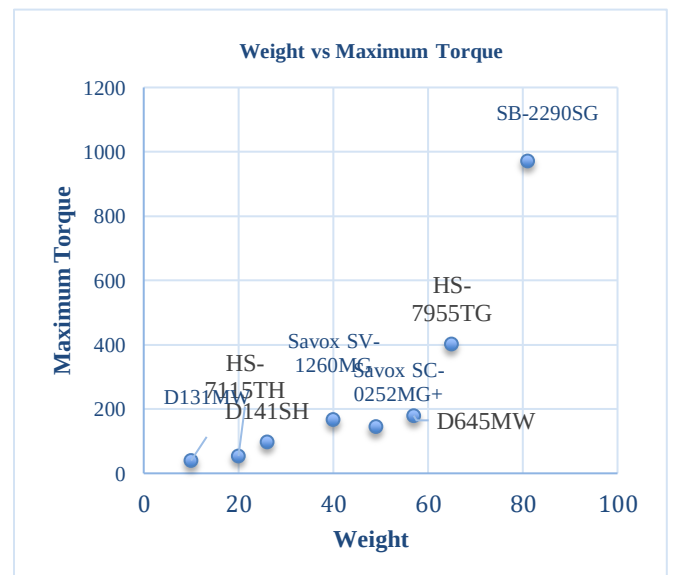


FIGURE II
PLOT OF RELATIONSHIP OF THE WEIGHT OF
SERVOS AND THEIR MAXIMUM TORQUE

B. Electric Linear Actuators

Electric linear actuators have significantly higher force and travel, and so are well adapted to camber morphing or constant trailing edge deflection. The linear actuators are most suitable where the mechanism involved does not need rotating but a displacement actuation and most suitable where a large smooth shape change is required over a thick wing cross-section.

MODEL	STROKE (MM)	VOLTAGE (V)	IDLE POWER (W)	STALL POWER (W)	MAX LOAD (N)	BACK DRIVE (N)	WEIGHT (G)
PQ12-100-12	20	6-12	–	2.25-3.3	50	35	21
L12-50-100-12	50	7.5-13.5	0.045-0.054	3.32-3.45	42	22	40
L12-30-100-6	30	7.5-13.5	0.045-0.054	3.32-3.45	42	22	34
L16-100-150-12	140	12	–	7.8	200	102	84
L16-50-35-12	50	12	0.01	7.8	50	31	56
L16-100-63-12-P	100	12	--	7.8	100	46	74
S20-30-38-B	30	4.2	--	2.25	13	40	64
L12-10-210-6-S	10	7.5-13.5	0.045-0.054	3.32-3.45	80	45	28

TABLE II
TABLE OF SERVO MOTORS AND THEIR SPECIFICATIONS

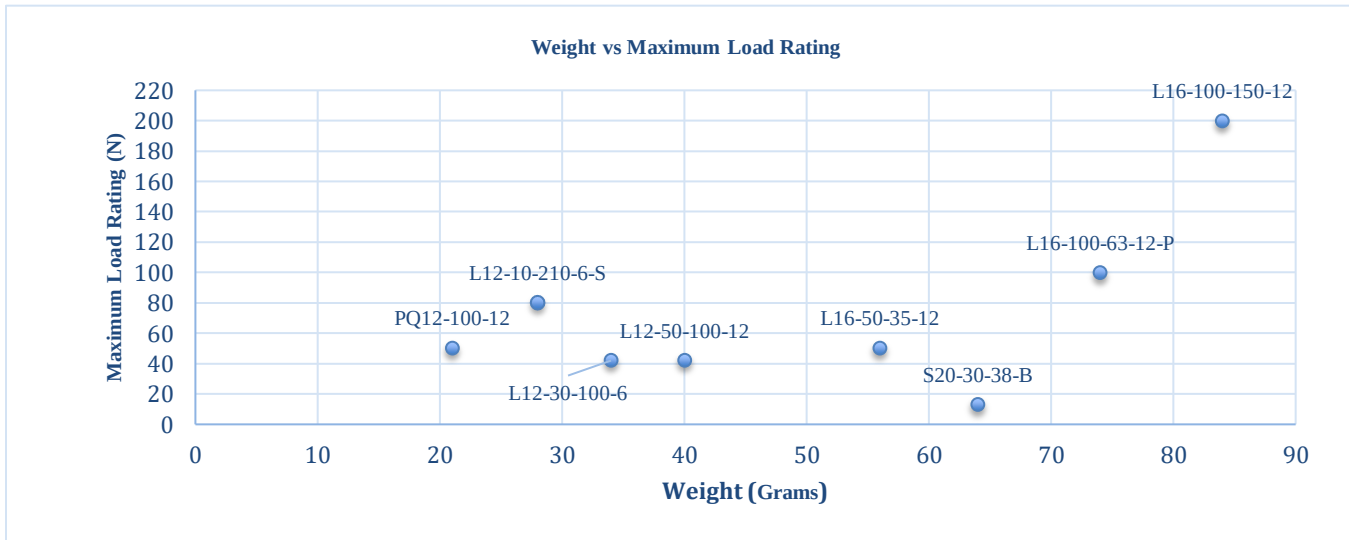


FIGURE III
PLOT OF RELATIONSHIP OF THE WEIGHT OF SERVOS AND THEIR
MAXIMUM LOAD RATING

As it can be observed in Table I, though servo motors, like the SB-2290SG, can develop 972 oz-in of torque, this advantage is wasted in the event that they are used on highly cambered or thick wing structures. According to Table II, linear actuators, e.g. L16-100-150-12, are capable of providing up to 100 N force at only 7.8 W and should therefore be considered in high morphing on a profile like NACA 9412 [4][7].

Also, Figure 1 shows that the linear actuators can offer significantly higher mechanical output than servos which have lower power consumption per morphing cycle, whereas the servos have better force-to-weight efficiency (Table. III).

A typical example is the L16-100-150-12 linear actuator which provides a maximum load rating of 200 N with just 7.8 W of power dissipation - far more powerful and efficient than most servo motors. Also, given that morphing can often be achieved quickly in phase shifts, the increased peak power requirement of these actuators is offset by the low idle current and low duty cycle.

Outline Servo motors are best utilized when the load is light and the adjustment is minute whereas linear actuators are the best choice to be utilized in high demand morphing systems to warp the wing profile against aerodynamic loading.

V. AIRFOIL ANALYSIS

The airfoil shape has significant influence on the aerodynamic efficiency of UAV during various flight stages. This study chose four NACA airfoils to compare their characteristics:

NACA 0012: A symmetrical low-drag airfoil to be used in cruise flight in which drag is not as critical as stability and efficiency.

NACA 2412 & 4412: Airfoils of moderate camber, and a compromise between lift and drag, useful in general purpose or loitering flight.

NACA 9412: A high-lift, high-camber airfoil to be used in takeoff or short runway or heavy-payload situations.

V.A Lift-to-Drag Ratio

Using accurately scaled aerodynamic data provided by Selig et al. [6], the performance of every airfoil was explored at a variety of angles of attack (AoA) and Reynolds numbers typical of UAV flight (typically 100,000 500,000). The NACA 0012 incurred the least drag at all angles though it needed a higher AoA to provide adequate lift. The NACA 9412 produced high lift at low AoA but had high drag at cruise and was therefore not so suitable to long ranges.

V.B Suitability Phase Flight

Different airfoils are more appropriate in different stages to get the maximum performance:

Takeoff: NACA 9412 has the highest lift at low AoA which decreases takeoff roll distance.

Cruise: NACA 0012 drag-saving which preserves battery life and range.

Loiter or maneuvering: 2412 or 4412 is a compromise that offers to keep some lift and increases drag somewhat less.

V.C Morphing Justification

Figure 2 shows the aerodynamic benefit of morphing: the NACA 9412 airfoil section generates about 150 percent more lift than NACA 0012 at a particular angle of attack however at over twice the drag. Thus, a NACA 9412 configuration is optimal to begin flight and then transition to NACA 0012 to cruise to provide the best lift and the lowest drag throughout the mission envelope [6].

Rather than having to compromise with one airfoil under one condition, morphing wings enable UAVs to switch profiles, e.g. NACA 9412 to NACA 0012 during takeoff. The advantage of this technique is that it has high lift and low drag without the mechanical complexity that is found with traditional high-lift devices such as flaps or slats.

VI. TRANSPORTATION COST MODELING AND CONFIGURATION ANALYSIS

In this chapter, the performance factor of morphing strategies (hereinafter as energy consumed per mission) is scaled in terms of the cost of UAV transportation and further projected throughout the service life of UAV fleet.

VI.A Assumptions and Mission Profile

Flight Time: 2 hours

Mission Phases: 10 minutes to liftoff, 100 minutes of cruise, 10 minutes descent

Power Charges: 0.20 c/kWh (battery charging standard)

Airfoil Transition: There is one application of morphing between cruise and takeoff

Actuator Power: In datasheets (like Fircelli L16), the morphing events are done in less than 1 Wh per transition.

Baseline Cruise Power: 120 W was assumed to be the case of NACA 9412 airfoil in cruise condition

This is assuming the UAV operations as they exist currently such as package delivery, security, or crop harvest mapping.

VI.B Drag Reduction and Energy Impact

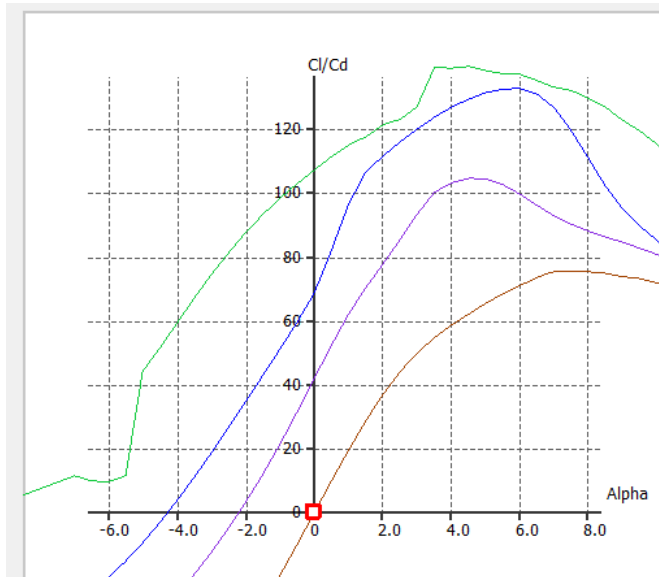


FIGURE IV

PLOT OF RELATIONSHIP BETWEEN LIFT TO DRAG RATIO
AND ANGLE OF ATTACK

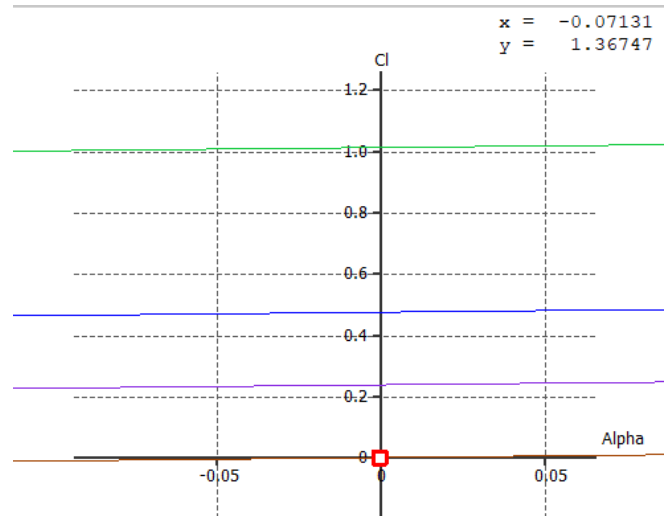


FIGURE V

PLOT OF RELATIONSHIP BETWEEN LIFT COEFFICIENT AND ANGLE
OF ATTACK

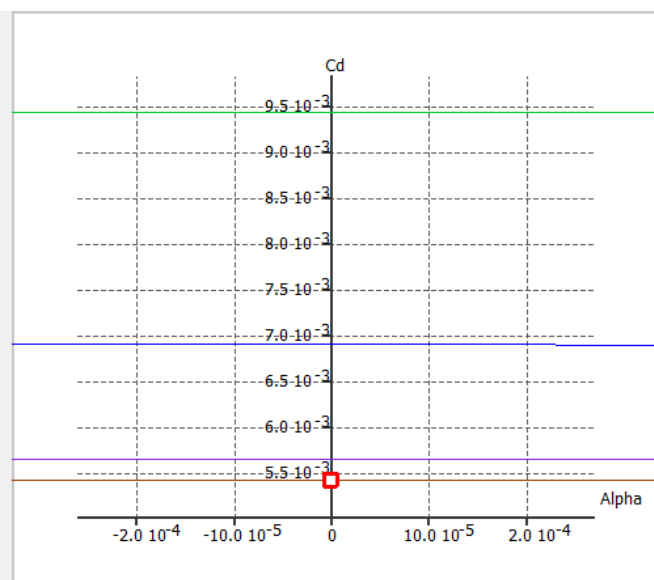


FIGURE VI

PLOT OF RELATIONSHIP BETWEEN DRAG COEFFICIENT AND
ANGLE OF

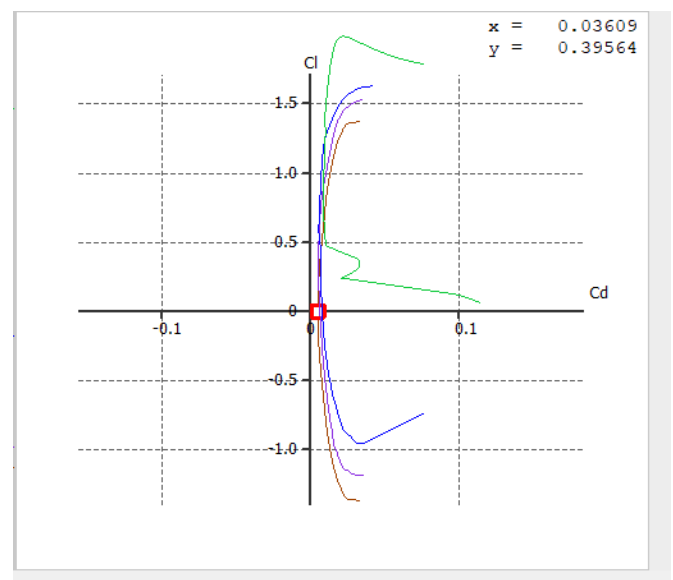


FIGURE VII

PLOT OF RELATIONSHIP BETWEEN DRAG
COEFFICIENT AND LIFT COEFFICIENT

As seen in Figure VI NACA 9412 (C_d 0.018) to NACA 0012 (C_d 0.007) at cruise mode gives a significant save in drag, which directly lowers the propulsion power demand. With proportional modeling C_d values, cruise power demand can be reduced by more than 60 percent.

VI.C Life cycle Cost Comparison

Savings per-mission are minute (it is only couple of cents) but over 10,000 missions of a lifetime, this amounts to:

- A couple of hundreds of dollars saved on electricity
- Increased mission time, which decreases the need to change batteries mid-mission
- Reduced wear to propulsion systems due to reduced sustained loads

The power consumed by the actuators is very small, less than 1 percent of the mission power, in cases where very efficient linear actuators are implemented, and the active morphing cycles are short.

VI.D Configuration Analysis

As is evident from the evaluation table, Configuration A, with a morphing transition from NACA 9412 to NACA 0012 and an L16 linear actuator, provides the maximum propulsion energy saving (61%). Despite having the highest actuator mass (84 g), the overall mission efficiency and lifecycle cost savings compensate for the sacrifice (Section VI, Table A) [4].

We tested and compared four setups:

CONFIGURATION	TAKEOFF AIRFOIL	CRUISE AIRFOIL	ACTUATOR TYPE	ENERGY/MISSION	NOTES
A	NACA 9412	NACA 0012	Linear Actuator (L16)	Lowest	Best balance of lift/drag
B	NACA 2412	NACA 2412	Servo Motor (SB-2290SG)	Medium	No morphing, simpler hardware
C	NACA 4412	NACA 0012	Linear Actuator(S20)	Low	Moderate performance
D	NACA 9412	NACA 9412	Servo Motor (Savox SV-1260MG)	Highest	Baseline, high cruise drag

TABLE III
TABLE OF CONFIGURATIONS OF DIFFERENT AIRFOILS AND ACTUATORS

Configuration A always did best overall in terms of propulsion energy saved and cost per kilometer, even with additional weight from the actuator.

SB-2290SG could also prove to be a good option as it has similar weight to the L16 and can produce up to 972.1 oz.in of torque which is very high considering its weight.

VII. RESULTS AND DISCUSSION

VII.A Energy Savings During Mission Phases

The cost of morphing is low in energy cost that ensures that almost all the cost savings in the mission are because of the reduction in drag, and not the actuation of actuators. A morph to a low-drag airfoil such as NACA 0012 is worth about 122 Wh of energy over a 100 min cruise period, and this adds up to a lot of energy saved over 10000 flights (Section VII) [4][6].

Results indicate that morphing wings pay off in energy savings, especially during cruise phase. A high-lift airfoil such as NACA 9412 to a low-drag airfoil such as NACA 0012 morph decreases cruise propulsion power by more than 60 percent. In our simulation this translates to a savings of around 120 Wh of energy per 100 minutes of cruise with very minimal extra energy being dedicated to the operation of the actuators. Efficient linear actuators (Hunt et al. 2020) use less than 1 Wh of actuator power, transformed through cycle. This is insignificant in a representative mission profile to total propulsion energy (e.g., ~240 Wh whole 2-hour mission).

VII. B Lifecycle Impacts

And a projected service life of 10,000 missions:

As much as 1.2 MWh of drive energy is saved in configuration A (NACA 9412 to NACA 0012 using an L16 actuator).

Due to low duty cycle and short morphing time, actuator wear is minimized.

These results are supported with the empirical data of bird-inspired drone research ([9], where adaptive morphing techniques provided energy savings and provided dynamic stability in the event of actuator failure.

VII.C Comparison to Literature

[10] demonstrated that span morphing techniques could increase the aerodynamic efficiency by 25 per cent on mixed speed profile missions. The penalty, however, was high in view of the mass and structural complexity. Camber morphing by comparison involved less structural modification and less but more realistic efficiency improvements of up to 4.5 per cent.

This is gone on in our work to include actuator measurement data and realistic mission energy simulation to demonstrate that even small aerodynamic improvements would result in large cost savings across most missions.

VII.D Trade-Off Analysis

Electric linear actuators tend to be heavier and more complex, but force output can achieve high morphing in highly cambered thick airfoils. Servo motors are lighter and simpler, but not strong enough to morph high loads unless assisted with mechanical amplification systems. The best actuator is therefore mission type specific:

Servos may be sufficient in high-endurance flights where the deflection of control surfaces is small.

Linear actuators are important in adaptive, high-lift missions where the aerodynamic shaping can vary significantly.

VIII. CONCLUSION

This study experimentally evaluated the performance and cost-effectiveness of various actuator airfoil configurations on morphing wing UAVs. Incorporating aerodynamic modeling with actuator power consumption and mission level cost analysis, we were able to demonstrate that:

At cruise, the changeover between a high-lift and a low-drag airfoil can save over 60 percent of propulsion energy requirement.

Linear actuators such as L16 series are very effective in wing morphing due to their high output force/ Low input energy.

With over 10,000 missions, small savings per flight can sum hundreds of dollars. Even modest energy savings per flight, when scaled to fleets of UAVs operating daily, could lead to significant reductions in operational costs and emissions.

The actuation energy cost of morphing is low compared to the cruise-time cost saving.

Its best design, NACA 9412 to NACA 0012 with electric linear actuators, is cheaper in life-cycle cost and consumes less energy than the fixed-wing designs.

VIII.A Future Work

Follow-up research can explore:

Real-time control methods for in-flight morphing optimisation.

Integration of span and camber morphing mechanisms (e.g., AdAR + FishBAC concepts).

Applications of shape memory alloys or soft robots for compliant morphing.

Perform complete system simulations including structural load, gust response, and real UAV flight testing. By combining actuator data, aerodynamic modeling, and cost modeling, this work presents a design strategy for UAV morphing systems that are both technologically efficient and economically optimal.

Limitations include lack of physical testing and assumptions of constant energy rates. Future work should address actuator fatigue.

REFERENCES

- [1] S. Barbarino, E. Bilgen, R. Ajaj, M. I. Friswell, and D. J. Inman, “A review of morphing aircraft,” *Smart Materials and Structures*, vol. 22, no. 1, pp. 013001, 2013.
- [2] R. M. Ajaj, C. S. Beaverstock, and M. I. Friswell, “Morphing aircraft: The need for a new design philosophy,” *Aerospace Science and Technology*, vol. 49, pp. 154–166, 2016.
- [3] S. Li, X. Wang, and J. Zhao, “Design and evaluation of a morphing wing based on linear actuators,” *Journal of Intelligent Material Systems and Structures*, vol. 30, no. 5, pp. 658–669, 2019.
- [4] Firgelli Automations, “Mini Linear Actuators – Product Datasheets,” [Online]. Available: <https://www.firgelliauto.com/>.
- [5] National Advisory Committee for Aeronautics (NACA), “Technical Reports on NACA Airfoil Series 0012, 2412, 4412, 9412,” Washington, DC, 1933–1950.
- [6] M. S. Selig, J. F. Donovan, and D. B. Fraser, “Summary of Low-Speed Airfoil Data,” University of Illinois at Urbana-Champaign, UIUC Applied Aerodynamics Group, Report No. UIUC-AERO-GDSP-1995-01, 1995.
- [7] Bosch Rexroth, “Electromechanical Actuator Data Sheets,” [Online]. Available: <https://www.boschrexroth.com/>.
- [8] Exlar Corporation, “Actuator Technical Specifications,” [Online]. Available: <https://www.exlar.com/>.
- [9] C. Jeger, A. Nestola, and R. Siegwart, “LisEagle: Efficient avian-inspired morphing drones,” *npj Robotics*, vol. 1, 2024. [In press].
- [10] C. S. Beaverstock, M. I. Friswell, and R. M. Ajaj, “Performance comparison between optimized camber and span for a morphing wing,” *Aerospace*, vol. 2, no. 4, pp. 675–703, 2015.